

Chenghao Li

Tel: (541)286-2464 | ash.morgan.le@gmail.com | Corvallis, OR | <https://lch9527.github.io/Li-s-Portfolio/>

RELEVANT TECHNICAL SKILLS

- **Graphics APIs / Engines:** Vulkan, OpenXR, Unreal Engine 5, Unity, OpenGL/OpenCL, GLSL/HLSL; C++, C, Python; Windows, Linux, Android.
- **GPU Performance:** RenderDoc, Unreal Insights, Quest profiling tools, CPU/GPU frame analysis, shader/material optimization, memory profiling, latency optimization.

PROFESSIONAL EXPERIENCE

Graphics Software Engineer

Feb 2025 – Current

Ylem Studio

Hustle (VR Game <https://www.ylemstudio.com/>)

- Led Quest 3 VR vertical slice in UE5, owning runtime systems, rendering performance constraints, and cross-discipline delivery with art, design, and gameplay developers.
- Built and optimized C++ gameplay/runtime systems for VR interaction, level flow, input, and physics assembly, with profiling-driven regression prevention.
- Profiled GPU/CPU frame behavior on mobile XR hardware; diagnosed rendering, asset, memory, and synchronization bottlenecks to sustain VR frame rate under Quest 3 thermal limits.
- Developed Vulkan/mobile rendering strategy, evaluating tonemapping, post-process cost, Mobile HDR tradeoffs, and shader/material complexity.

Unreal Engine Developer

July 2023 - Jan 2025

Gamma Interactive

Remote

Published title: X8 (VR Game <https://www.meta.com/experiences/4518172508241377/>)

- Implemented UE gameplay and visual effects support, including volumetric fog integration/modification and throwing-input systems for a shipped VR title.
- Investigated and fixed multiplayer/runtime bugs involving replicated gameplay state, input handling, throwing interactions, and character behavior.
- Optimized Quest runtime performance through frame captures, gameplay/rendering overhead reduction, and validation on target VR hardware.

US Navy Aircraft Carrier Simulation project (VR/mobile)

- Built a Unity VR/mobile carrier-deck simulation enabling multiplayer collaboration on helicopter cargo drops and F-35 catapult-launch workflows.

Macht(VR Game <https://playmacht.com/>)

- Built a custom VR movement component supporting server correction and multiplayer compatibility.
- Created gameplay-reactive character shader effects and material controls.

Research Assistant

May 2021 - Aug 2023

NVIDIA Graphics & Image Technology Laboratory

Corvallis

- Secured EDA (U.S. Department of Commerce) funding (award #07-79-07914) for an interactive VR forest simulation system, following a successful demonstration to the Oregon Department of Forestry.
- Implemented a VR education solution, using it in university classes and enhancing 200+ civil engineering students' grasp of building structures and force distribution, with NVIDIA Omniverse.

SELECTED PROJECTS

Vulkan Ray-Tracing (<https://github.com/lch9527/Vulkan>)

February 2023

- Built a Vulkan RTX ray tracer in C++/GLSL with VK_KHR_ray_tracing pipeline, acceleration structures, SBT setup, raygen/miss/closest-hit shaders, validation layers, and RenderDoc debugging.

EDUCATION

Bachelor of **Computer Science**, Sept 2019 - June 2023

Oregon State University, Major GPA: 3.65

Corvallis, OR